

TOLL9 (Q5TWL7, AGAP006974) — Ligand and Immune-Pathway Specificity: Final Curation Report

Target gene: *Anopheles gambiae* TOLL9 (UniProt Q5TWL7 / AGAP006974) **Focus:** function_assignment — toll9-ligand-and-immune-pathway-specificity **Source:** genes/ANOGA/TOLL9/TOLL9-ai-review.yaml

Executive Judgment

Verdict: Partially supported / the strong claim is over-annotated as a direct function assignment.

The seed hypothesis proposes that *A. gambiae* TOLL9 (1) **has pattern-recognition receptor (PRR) activity** and (2) **participates in innate immunity through the Toll signaling pathway**. After three iterations of orthology, domain-architecture, partner-availability, and literature analysis, the defensible position is nuanced:

- **What is supported (retain):** TOLL9 is a canonical Toll/TLR-family transmembrane receptor. Its generic, phylogeny-transferred annotations — **signaling receptor activity (GO:0038023, MF)**, **signal transduction (GO:0007165, BP)**, and **plasma membrane (GO:0005886, CC)** — are consistent with its domain architecture (LRR ectodomain + two TM helices + cytoplasmic TIR) and with the ancestral scope of PANTHER family PTHR24365. These should be retained.
- **What is NOT supported as a direct function assignment (flag as leads, do not assert):** The specific claims that TOLL9 **directly has PRR activity (GO:0038187)** and functions as an **innate-immune Toll-signaling receptor** are biologically plausible but **experimentally unproven in *Anopheles gambiae***. The only direct PRR/LPS evidence comes from the more distant lepidopteran *Bombyx mori*, and even there the downstream signaling was demonstrated using a **chimeric receptor whose transmembrane and TIR regions are mouse TLR4** — so the native insect signaling readout was never shown. The closest genetic

test, a *Drosophila* Toll-9 loss-of-function mutant (~36% identical to the target), shows **no antibacterial defect**. The *Anopheles* mosGILT data are **correlative expression changes, not a perturbation of TOLL9 itself**.

Most important caveat: A "small experimental donor set" is not automatically weak evidence — but in this case the donor evidence is both **single-species** and **mechanistically partial** (chimeric TM/TIR), and the closest ortholog contradicts a required antibacterial role. That combination is exactly what should prevent a species-level MF PRR assignment while leaving the immune hypothesis open as a curator lead.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence limitation
UniProt Q5TWL7 (database)	Computational / database	Qualifies	What is currently annotated and by what evidence	Protein existence "Inferred from homology" (level 3). All four GO terms are IBA (phylogenetic, GO_Central): GO:0005886, GO:0038023, GO:0007165, GO:0006954. No GO:0038187 (PRR) or Toll-signaling/defense BP term is transferred.	<i>A. gambiae</i> , full proteome record	High for the annotation state; no experimental evidence for this protein
Domain analysis (this work)	Structural / computational	Supports (generic)	Is TOLL9 a canonical Toll/TLR receptor	LRR ectodomain (Pfam PF13855 LRR_8 ×2; SMART LRR_TYP ×7) + two TM helices (aa 29–49, 607–631) + cytoplasmic TIR (aa 667–807; PF01582; PROSITE PS50104). Family PTHR24365 ("TOLL-LIKE RECEPTOR 1"),	Sequence-derived	High for the architecture; architecture alone does not specify ligand or PRR activity

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence and limitation
				subfamily SF530.		
PMID: 33963082	Direct assay (heterologous/chimeric)	Qualifies / partial support	Does insect Toll9 recognize LPS via MD-2-like partners	BmToll9 recognizes LPS through two MD-2-like partners (BmMD-2A/B). Signaling competence shown with a chimera = BmToll9 ectodomain + mouse TLR4 TM + TIR , activating LPS-induced inflammatory-factor release in murine cells only with BmMD-2A/B present.	<i>Bombyx mori</i> ectodomain; murine cell readout	Medium; no insect TM/ and an insect signaling readout was NOT demonstrated in single domain species
PMID: 21386906	Mutant phenotype (loss-of-function)	Refutes (required immune role)	Is Toll-9 required for antibacterial defense	Complete <i>Drosophila</i> Toll-9 LOF allele shows no defect in basal or infection-induced antibacterial (AMP) response; authors acknowledge overexpression-	<i>D. melanogaster</i> , whole animal	High for the tested conditions; does not exclude context-specific or non-AMP roles

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence and limitation
				induced signaling.		
PMID: 38191283	Expression correlation (RNA-seq)	Qualifies (weak)	Is AGAP006974 immune-regulated in mosquito	AGAP006974 is induced in mosGILT ^Δ null <i>A. gambiae</i> ; co-regulation of immune genes.	<i>A. gambiae</i> , transcriptome	Low for causation; TOLL9 itself was not perturbed; correlation only
PMID: 35238667	Review-level	Qualifies	Is the Toll9/MD-2/LPS property established	States Toll9/MD-2-like complex in silkworm acts as PRR recognizing LPS.	Review	Medium; reiterates single-species <i>Bombyx</i> findings
PMID: 25573502	Expression/ RNAi (paralog/ related clade)	Competing / context	Do Toll9-clade receptors regulate AMPs elsewhere	Crayfish PcToll (clusters with <i>Drosophila/ Anopheles</i> Toll9 and human Tolls) is bacteria-inducible; RNAi reduces crustins/ALF2/ Lys1; S2 overexpression induces AMPs.	<i>Procambarus clarkii</i>	Low-medium confidence for transfer to different species; overexpression RNAi, not mosquito ortholog
MD-2 partner search (this work)	Computational / database	Supports feasibility	Are MD-2-like partners available in <i>Anopheles</i>	≥16 ML-domain proteins (Pfam PF02221 / IPR003172) in taxid 7165 (e.g., Q7QJ15,	<i>A. gambiae</i> proteome	High for presence; demonstrated TOLL9 interaction

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence limitation
				Q7PR71, Q7QCX7, cluster Q7QCX4/5/7/8/9).		
Ortholog alignment (this work)	Structural / evolutionary	Qualifies	How close is the target to the LOF-tested ortholog	Needleman–Wunsch global alignment Q5TWL7 (812 aa) vs <i>Dmel</i> Toll-9 Q9VPH1 (900 aa) = 35.6% identity (343/963). Single-copy in both; same PTHR24365 subfamily.	Dipteran comparison	High for the number; 3 identity m LOF is rele but transfe imperfect

Key Findings (expanded)

F001 — TOLL9 carries only generic receptor annotations, transferred phylogenetically

UniProt Q5TWL7 has protein existence "Inferred from homology" (level 3) — there is **no experimental characterization of this specific protein**. Its domain architecture is unambiguously that of a Toll/TLR receptor: an extracellular leucine-rich-repeat (LRR) ectodomain (Pfam PF13855 LRR_8 ×2; SMART LRR_TYP ×7), two transmembrane helices (aa 29–49 and 607–631), and a cytoplasmic TIR signaling domain (aa 667–807; Pfam PF01582; PROSITE PS50104). It belongs to PANTHER PTHR24365 ("TOLL-LIKE RECEPTOR 1"), subfamily SF530. All four of its GO annotations are **IBA** (phylogenetically inferred via GO_Central): plasma membrane (GO:0005886), signaling receptor activity (GO:0038023), signal transduction (GO:0007165), and inflammatory response (GO:0006954). Notably **absent** are GO:0038187 (pattern recognition receptor activity) and any Toll-signaling or defense-response BP term. The seed hypothesis's stronger claims are therefore **not** currently supported by the transferred

annotation set — they would be new assertions. The current leaf PTN000688561 in PTHR24365 descends from PTN002808045, which supports only generic receptor/signaling/plasma-membrane properties — consistent with the observed transfer scope.

F002 — Direct PRR/LPS evidence is *Bombyx*-only and mechanistically chimeric

The direct experimental basis for insect Toll9 as an LPS-recognizing PRR is [PMID: 33963082](#): BmToll9 (*Bombyx mori*) recognizes LPS through two MD-2-like partners (BmMD-2A/BmMD-2B). However, the demonstration of downstream signaling used a **chimeric receptor** in which the BmToll9 ectodomain was fused to **mouse TLR4 transmembrane and TIR domains**, and the readout (LPS-induced inflammatory-factor release) occurred in **murine cells**. Thus the insect ectodomain's LPS/MD-2 binding is real, but native insect TM/TIR signaling and an insect functional readout were not established. Against a required immune role, the *Drosophila* Toll-9 complete loss-of-function mutant ([PMID: 21386906](#)) shows no defect in basal or infection-induced AMP responses, while acknowledging overexpression-induced signaling. The *Anopheles* mosGILT^{null} data ([PMID: 38191283](#)) are transcriptional co-regulation only — TOLL9 itself was never perturbed.

F003 — MD-2-like partners exist in *Anopheles*, so partner-absence does not block the mechanism

A UniProt search (taxid 7165; Pfam PF02221 ML domain / IPR003172 MD-2-related lipid-recognition) returns ≥ 16 *A. gambiae* proteins (e.g., Q7QJ15, Q7PR71, Q7QCX7, A7USS7, A7UVK2, F5HMB6, and a cluster Q7QCX4/5/7/8/9 of ~148–170 aa). These are the mosquito counterparts of *Bombyx* BmMD-2A/B and mammalian MD-2. The presence of this accessory class means the *Bombyx*-style mechanism is **not excluded** in *Anopheles* on partner-availability grounds. However, **none** is annotated as interacting with TOLL9/AGAP006974, and no LPS-binding assay ties any of them to the target — so this is feasibility, not evidence of function.

F004 — *Anopheles* TOLL9 vs *Drosophila* Toll-9 are ~36% identical

A Needleman–Wunsch global alignment (match +1 / mismatch –1 / gap –1) of full-length *A. gambiae* TOLL9 (Q5TWL7, 812 aa) against *D. melanogaster* Toll-9 (Q9VPH1, 900 aa) yields **35.6% identity** (343 identical positions across 963 aligned columns). Both are single-copy members of the same PANTHER family/subfamily and share the LRR+TM+TIR architecture. This level of divergence is high enough that the dipteran LOF-negative is the **most relevant genetic**

evidence available, yet imperfect enough that it cannot fully exclude a divergent function. (The *Bombyx* BmToll9 could not be resolved to a single reviewed UniProt accession, so an *Anopheles*-vs-*Bombyx* identity was not computed — a stated limitation.)

Mechanistic Model / Interpretation

The immediate molecular function under test is **direct microbial-PAMP recognition (PRR activity) coupled to Toll-pathway signal transduction**. The distinction that governs the curation call is:

Direct gene-product activity		Downstream / inferred (NOT direct)
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- Transmembrane receptor topology	YES	- AMP induction phenotypes (<i>Drosophila</i> LOF = none)
- TIR-domain signaling capacity	~	- Immune-gene co-regulation (<i>mosGILT</i> RNA-seq)
- LPS/MD-2 recognition (<i>Bombyx</i> , chimeric TM/TIR)	~	- "Innate immunity" as a pathway outcome
- LPS recognition in <i>Anopheles</i>	NO	- Inflammatory response (vertebrate-only, GO:0006954)

Three mechanistic separations drive the verdict:

- 1. Recognition of a processed host cytokine vs. microbial PAMP.** Canonical *Drosophila* Toll binds the cleaved cytokine **Spätzle**, not microbes directly. The seed hypothesis correctly warns against assuming every insect Toll uses Spätzle *and* against equating host-cytokine recognition with GO:0038187 microbial-PAMP recognition. TOLL9's actual ligand in *Anopheles* is **unknown**.
- 2. Ectodomain binding vs. native signaling.** In *Bombyx*, the ectodomain's LPS/MD-2 binding is demonstrated, but the productive signaling event used **mouse TLR4 TM+TIR**. This means the insect protein's own TM/TIR was not shown to transduce the signal — a critical gap when transferring an MF PRR term.
- 3. Correlation vs. perturbation.** The *Anopheles* *mosGILT*^{null} induction of AGAP006974 is transcriptional co-regulation; TOLL9 was never knocked down or out. It cannot establish direct function.

Synthesizing: TOLL9 is architecturally a receptor, and the raw ingredients for an *Anopheles* Toll9/MD-2/LPS PRR complex exist (partner class present). But the chain of direct evidence for **this species** is broken at every functional link — no ligand assay, no partner interaction, no

loss-of-function phenotype — and the closest genetic ortholog argues against a required antibacterial role. The proper model is a **plausible-but-unproven immune receptor** whose only defensible current annotations are generic.

GO Curation Implications

Lead recommendation (requires curator verification):

GO term	Aspect	Current status	Recommended action	Rationale
GO:0038023 signaling receptor activity	MF	IBA present	Retain	Matches domain architecture and ancestral PTHR24365 scope
GO:0007165 signal transduction	BP	IBA present	Retain	Consistent with TIR-domain receptor
GO:0005886 plasma membrane	CC	IBA present	Retain	Two TM helices; canonical Toll topology
GO:0038187 pattern recognition receptor activity	MF	Absent	Do NOT add as an assertion; flag as lead	Direct PRR/LPS evidence is <i>Bombyx</i> -only and chimeric; no <i>Anopheles</i> assay
Toll signaling pathway / defense response BP	BP	Absent	Do NOT add; flag as lead	Closest ortholog (<i>Drosophila</i>) LOF-negative; mosquito data correlative
GO:0006954 inflammatory response	BP	IBA present	Out of scope here (resolved separately by vertebrate-only GO:0006954 constraint)	Not part of this question

The evidence does not justify promoting TOLL9 from a **generic signaling receptor** to a **specific microbial-PAMP pattern-recognition receptor** at the species level. "Protein binding" is explicitly avoided as a recommendation; the informative and defensible MF term remains **signaling receptor activity (GO:0038023)**.

Evidence Base

- **PMID: 33963082** — *Toll9 from Bombyx mori functions as a pattern recognition receptor sharing features with TLR4.* Establishes the LPS/MD-2-like PRR property for insect Toll9 and situates Toll9 with the mammalian TLR4 group. Verified snippet: "*Toll9 members differ from other insect Tolls by clustering with the mammalian TLR4 group, which recognizes lipopolysaccharide (LPS) through interaction with myeloid differentiation-2 (MD-2)-like proteins.*" Crucially, the functional signaling assay used a chimeric receptor with mouse TLR4 TM+TIR, so it demonstrates ectodomain ligand recognition but not native insect signal transduction.
 - **PMID: 21386906** — *Lack of an antibacterial response defect in Drosophila Toll-9 mutant.* Loss-of-function genetics in the closest dipteran ortholog directly weakens a required antibacterial immune role. Verified snippet: "*Our results suggest that Toll-9 is neither required to maintain a basal anti-microbial response nor to mount an efficient immune response to bacterial infection.*"
 - **PMID: 35238667** — Review-level statement confirming the Toll9/MD-2/LPS PRR property is a *Bombyx*-specific experimental finding: "*the Toll9/MD-2-like protein complex in the silkworm, Bombyx mori, acts as an innate pattern-recognition receptor that recognizes lipopolysaccharide (LPS).*" Reinforces that this is a single donor species.
 - **PMID: 38191283** — Reports AGAP006974 induction after mosGILT disruption in *A. gambiae* (RNA-seq). This is expression correlation, not a TOLL9 perturbation, and cannot establish direct function.
 - **PMID: 25573502** — Crayfish PcToll (clusters with *Drosophila/Anopheles* Toll9 and human Tolls) is bacteria-inducible; RNAi and S2 overexpression implicate it in AMP regulation. A competing/context line of evidence from a distant species that shows the Toll9 clade *can* be immune-associated, but does not establish direct PRR activity for the mosquito ortholog.
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Conflicts and Alternatives

- **Paralog/clade confusion:** Cross-species Toll9-clade immune associations (crayfish PcToll, PMID: 25573502) are suggestive but come from distant taxa and overexpression/RNAi, not the mosquito ortholog. They should not be transferred as direct evidence.
 - **Organism-specific divergence:** The strongest genetic test (*Drosophila* LOF) is negative for antibacterial defense, and the strongest biochemical test (*Bombyx*) is chimeric and single-species. These pull in opposite directions and together argue for a **species-level "unresolved/lead"** status rather than assertion.
 - **Correlation $\neq$ causation:** mosGILT^{null} induction of AGAP006974 is a co-regulation signal, easily produced by generic stress/immune transcriptional programs without TOLL9 having direct PRR activity.
 - **Database carry-over risk:** Adding GO:0038187 by phylogeny would propagate a *Bombyx*-specific, chimeric-signaling finding across the family — precisely the over-annotation the review should avoid.
 - **Ligand-class ambiguity:** Whether TOLL9's ligand is a microbial PAMP ($\rightarrow$ GO:0038187) or a processed host cytokine (Spätzle-like) is unknown; conflating the two would mis-annotate the MF.
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Limitations and Knowledge Gaps

1. **No *Anopheles*-specific functional data.** There is no LPS/PAMP binding assay, no ligand identification, no interaction assay with any MD-2-like partner, and no loss-of-function phenotype for AGAP006974. Everything about TOLL9's ligand and immune role in *Anopheles* is inferred. *What would resolve it:* direct LPS/microbial-binding assay and RNAi/CRISPR knockdown with infection challenge.
2. **Chimeric-signaling caveat unresolved for any insect.** Even the strongest donor (*Bombyx*) did not show native insect TM/TIR signaling. *What would resolve it:* reconstitute full-length insect Toll9 signaling in an insect cell system with an insect readout (e.g., Drosomycin-luciferase in S2 cells).

3. **Bombyx–Anopheles distance not quantified.** BmToll9 could not be mapped to a single reviewed accession, so the identity to the actual donor of the experimental property was not computed. *What would resolve it:* obtain the exact BmToll9 sequence from PMID33963082 and align.
 4. **Partner interaction unproven.** ≥ 16 ML-domain proteins exist in *Anopheles*, but which (if any) partners TOLL9 is unknown. *What would resolve it:* co-IP / pull-down or AlphaFold-Multimer screening of the TOLL9 ectodomain against the ML-domain set.
 5. **Ligand identity.** Microbial PAMP (GO:0038187) vs. processed host cytokine (Spätzle-like) is undetermined — these map to different GO terms. *What would resolve it:* ligand pull-down and Spätzle-processing assays.
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Discriminating Tests

1. **RNAi / CRISPR knockdown of AGAP006974 in *A. gambiae*** followed by bacterial (Gram-negative/LPS) and fungal challenge, measuring AMP induction and survival — directly tests the immune/Toll-pathway claim in the target species.
 2. **Direct LPS/PAMP binding assay** with recombinant TOLL9 ectodomain $\pm$ *Anopheles* ML-domain partners — tests GO:0038187 PRR activity directly.
 3. **Native insect signaling reconstitution** (full-length TOLL9, not a mouse-TLR4 chimera) in S2 cells with a Drosomycin/AMP reporter — resolves the chimeric-TM/TIR caveat.
 4. **TOLL9–ML-domain interaction screen** (co-IP or AlphaFold-Multimer across the ≥ 16 candidates) — tests the accessory-partner requirement.
 5. **Ligand-class discrimination:** microbial PAMP vs. processed Spätzle-like cytokine — determines the correct GO term (GO:0038187 vs. cytokine-receptor terms).
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Proposed Follow-up Actions / Curation Leads (require curator verification)

Action changes: - **Retain** generic IBA annotations: GO:0038023 (signaling receptor activity, MF), GO:0007165 (signal transduction, BP), GO:0005886 (plasma membrane, CC). - **Do NOT assert** GO:0038187 (PRR activity) or Toll-signaling/defense BP terms as species-level

annotations. Record them as **hypothesis leads** with the *Bombyx* chimeric caveat and the *Drosophila* LOF-negative explicitly noted. - GO:0006954 (inflammatory response) is out of scope here (already resolved by the vertebrate-only constraint).

Candidate references with snippets to verify: - [PMID: 33963082](#): "Toll9 members differ from other insect Tolls by clustering with the mammalian TLR4 group, which recognizes lipopolysaccharide (LPS) through interaction with myeloid differentiation-2 (MD-2)-like proteins" — supports PRR/LPS property, but note chimeric TM/TIR. - [PMID: 21386906](#): "Toll-9 is neither required to maintain a basal anti-microbial response nor to mount an efficient immune response to bacterial infection" — refutes required antibacterial role in the closest ortholog. - [PMID: 35238667](#): confirms the PRR property is *Bombyx*-specific (single donor).

Suggested curator questions: - Is a chimeric-receptor demonstration (mouse TLR4 TM+TIR) sufficient to transfer an MF PRR term to a distant ortholog? (Recommended answer: no.) - Should GO:0038187 be added only after an *Anopheles*-specific binding or LOF assay exists? (Recommended: yes.)

Suggested experiments: as listed under Discriminating Tests.

Summary

A. gambiae TOLL9 is a bona fide Toll/TLR-family transmembrane receptor whose **generic** annotations — signaling receptor activity (GO:0038023), signal transduction (GO:0007165), and plasma membrane (GO:0005886) — are well justified by domain architecture and the ancestral PTHR24365 transfer scope and should be **retained**. The seed hypothesis's stronger claims — that TOLL9 **directly has pattern-recognition receptor activity (GO:0038187)** and functions as an **innate-immune Toll-signaling receptor** — are **biologically plausible but experimentally unproven in this species**. The direct PRR/LPS evidence is confined to the more distant *Bombyx mori* and was demonstrated through a receptor chimera bearing mouse TLR4 transmembrane and TIR domains; the closest genetic test (*Drosophila* Toll-9 LOF, ~36% identical) shows no antibacterial defect; and the *Anopheles* mosGILT signal is merely correlative. The MD-2-like accessory partner class does exist in *Anopheles* (≥ 16 ML-domain proteins), so the mechanism is not excluded — but no TOLL9 ligand, interaction, or loss-of-function phenotype has been demonstrated. The correct curation posture is therefore to **retain the generic receptor terms and treat PRR/immune terms as leads, not supported assignments**.

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